

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester IV Examination, 2022-23

Paper No. BPT-401: Electronics and Modern Physics

Time: 3 Hours

Maximum Marks: 70

*Instructions: Answer **five** questions including Question 1 which is compulsory. Symbols have their usual meanings.*

Question 1. (Compulsory — Answer any five parts, 4 marks each.)

- (a) What are the differences between metals, insulators, and semiconductors? Describe them on the basis of the band theory of solids. [4]
- (b) Define the phase velocity and group velocity of matter waves. Obtain the relation between them. [4]
- (c) If a dielectric is introduced between the plates of a parallel plate capacitor, show how the induced charge varies as a function of the dielectric constant. Show that the dielectric constant approaches infinity for a metal. [4]
- (d) Differentiate between the electronic, ionic, and orientational polarizabilities of a dielectric material. [4]
- (e) For an intrinsic semiconductor, the effective masses of hole and electron are $0.36 m_0$ and $0.54 m_0$ respectively (m_0 being the rest mass of an electron). Find the position of the Fermi level in this semiconductor at $T = 300$ K. [4]
- (f) Light of wavelength 400 nm is incident on a metal surface with a work function of 2.5 eV. Calculate the maximum kinetic energy of the emitted photoelectrons. [4]
- (g) How is the current amplification factor α different from the current gain β ? Determine the current gain β and emitter current I_E for a transistor operating in CE configuration, where $I_B = 50 \mu\text{A}$ and $I_C = 3.65 \text{ mA}$. [4]

Question 2.

- (a) Derive the expression for the local molecular field inside a spherical cavity of a dielectric material and establish the Clausius-Mossotti equation. What are the limitations of this equation? [7]
- (b) Explain the three electric vectors: electric field intensity $\vec{E}$, dielectric polarization $\vec{P}$, and electric displacement $\vec{D}$. Establish the relation among them. [7]

OR

- (a) Derive the Langevin-Debye formula for the dielectric susceptibility of a polar medium. [7]

- (b) Describe the formation of the depletion layer in an unbiased p-n junction. Sketch the barrier potential and show that the barrier potential is given by:

$$V_0 = \frac{kT}{q} \ln \left(\frac{N_a N_d}{n_i^2} \right)$$

where the symbols have their usual meanings.

[7]

Question 3.

- (a) What is meant by the peak inverse voltage (PIV) of a rectifier? Prove that the regulation for both half-wave and full-wave rectifiers is given by the ratio of forward diode resistance to the load resistance. [5.5]
- (b) Explain the difference between the Zeeman effect and the Compton effect. Give a brief account of Einstein's explanation of the photoelectric effect on the basis of quantum theory. [5]
- (c) Show that the de Broglie wavelength (λ) for a material particle of rest mass m_0 and charge q accelerated from rest through a potential difference of V volts relativistically is given by:

$$\lambda = \frac{h}{\sqrt{2m_0 qV \left(1 + \frac{qV}{2m_0 c^2} \right)}}$$

where c is the speed of light.

[3.5]

OR

- (a) "Zener breakdown voltage has a negative temperature coefficient whereas the avalanche breakdown voltage has a positive temperature coefficient." Explain this statement. How does a Zener diode work as a voltage regulator? [7]
- (b) Derive the expression for the change in wavelength of a photon due to Compton scattering with a stationary electron. Explain the significance of the Compton wavelength. [7]

Question 4.

- (a) State Bragg's law of X-ray diffraction in crystals. Describe the Bragg X-ray spectrometer method for determining the wavelength of X-rays. [5]
- (b) What is the meaning of biasing a transistor? Discuss the different methods used for transistor biasing and provide a neat circuit diagram for a self-biased transistor. [6]
- (c) Explain the terms "polar vector" and "axial vector" with one example of each. [3]

OR

- (a) A silicon n-p-n transistor having $\beta = 100$, $I_{CO} = 22 \text{ nA}$ is operated in CE configuration. Assuming $V_{BE} = 0.7 \text{ V}$ and collector supply voltage $V_{CC} = 12 \text{ V}$ with collector load resistance $R_C = 2 \text{ k}\Omega$, determine the transistor currents and the region of operation. What happens if R_C is indefinitely increased? [10]

- (b) What is a load line? Explain using a suitable diagram how the DC load line can be constructed for a common emitter transistor amplifier. [4]
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Question 5.

- (a) Describe the operation of an R-C coupled common-emitter transistor amplifier and explain its frequency response curve. [7]
- (b) What is the Early effect in transistors? Explain why the transistor action is lost if the base width is reduced to zero. [7]

OR

- (a) Show that the energy lost by a photon in Compton scattering with a free stationary electron can be written as:

$$\Delta E = \frac{(h\nu)^2(1 - \cos \theta)}{m_0c^2 + h\nu(1 - \cos \theta)}$$

where m_0 is the rest mass of the electron, $h\nu$ is the energy of the incident photon, and θ is the scattering angle of the photon. [7]

- (b) Calculate the maximum percentage of energy that can be lost by a 1 MeV photon in a collision with a stationary electron (rest mass energy of electron is 0.51 MeV). [7]